K L UNIVERSITY

FRESHMAN ENGINEERING DEPARTMENT

A Project Based Lab Report
On
Lost item logger system for college campus

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DEPARTMENT OF BASIC ENGINEERING SCIENCES

CERTIFICATE

This is to certify that the project based laboratory report entitled "Lost Item Logger System for college campus" bearing to the **Department of Basic Engineering Sciences-1 , KL University** in partial fulfillment of the requirements for the completion of a project based Laboratory in " Python Programming (25SC2007)" course in I B Tech I Semester, is a Bonafede record of the work carried out by him/her under my supervision during the academic year 2025 – 2026.

PROJECT SUPERVISOR

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ABSTRACT

The Lost Item Logger System is a comprehensive digital platform developed to address the recurring issue of misplaced belongings within a college campus. Traditional methods of reporting lost items, such as handwritten logs or verbal communication, often lead to inefficiency, miscommunication, and low recovery rates. To overcome these challenges, the system introduces a structured, technology-driven approach that enables students, faculty, and administrative staff to record, track, and retrieve lost or found items in a centralized manner. By integrating modern database management and intuitive user interfaces, the system ensures that every report is systematically stored, easily searchable, and accessible to authorized users, thereby reducing confusion and enhancing the overall campus experience.

The system's functionality extends beyond simple logging by incorporating advanced features such as categorization of items, location tagging, and automated notifications to potential owners. Users can input detailed descriptions, upload images, and specify the time and place of loss or discovery, which significantly improves the accuracy of matching items to their rightful owners. Administrators play a crucial role in verifying entries, preventing fraudulent claims, and maintaining the integrity of the database. Additionally, the system fosters a sense of accountability and community responsibility by encouraging students and staff to actively participate in the recovery process. From a technical perspective, the Lost Item Logger System is designed with scalability and security in mind, ensuring that sensitive information is protected while accommodating the growing needs of a dynamic campus environment. Beyond its immediate utility, the system contributes to building a culture of trust and cooperation, minimizing the stress associated with lost belongings, and allowing individuals to focus more effectively on their academic and professional goals. In essence, it transforms a common campus problem into an opportunity for technological innovation and community engagement, making the college environment more organized, efficient, and student-friendly.

Key Words: Lost Items, Campus System, Digital Logger

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INTRODUCTION

The Lost Item Logger System is designed to provide a structured and efficient solution for managing lost and found items within a college campus. In traditional settings, students and staff often rely on manual registers or informal communication, which leads to delays, confusion, and low recovery rates. This system introduces a centralized digital platform where users can log details of lost or found items, search through a secure database, and receive notifications when potential matches are identified. By integrating technology into this everyday challenge, the system not only improves recovery rates but also fosters accountability, enhances campus organization, and strengthens the sense of community among students and staff.

The importance of such a system lies in its ability to bridge the gap between individuals who lose items and those who find them, ensuring that valuable belongings are not permanently misplaced. With the increasing reliance on digital solutions in academic institutions, the Lost Item Logger System aligns with modern campus infrastructure by offering accessibility through computers and mobile devices. It reduces administrative burden, minimizes human error, and provides transparency in the recovery process. Furthermore, the system encourages responsible behavior among students and staff, as it creates a culture where reporting and returning found items becomes a shared responsibility, ultimately contributing to a safer and more trustworthy campus environment.

- Centralized platform for logging lost and found items
- Searchable database with detailed item descriptions
- Notification system to alert users of potential matches
- Administrator verification to ensure accuracy and prevent misuse
- Promotes accountability and community trust within the campus

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AIM

The primary aim of the Lost Item Logger System is to develop a centralized, technology-driven platform that simplifies and streamlines the process of reporting, tracking, and recovering lost and found items within a college campus. The system seeks to eliminate inefficiencies associated with traditional manual methods by providing a secure, user-friendly interface where students and staff can log item details, search through a structured database, and receive timely notifications about potential matches. By ensuring accuracy, transparency, and accountability, the system aims to foster a more organized and trustworthy campus environment, ultimately reducing stress and enhancing the overall academic and social experience of the college community.

The modules in this project are -

→User Module

→ Admin Module

→ Database Module

→Notification Module

→ Security Module

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ADVANTAGES

- Efficient recovery of lost items through a centralized digital platform
- Transparent records with admin verification to reduce misuse and confusion
- Strengthened cooperation and trust among students and staff in the campus community

DISADVANTAGES

- Requires regular internet connectivity for smooth functioning
- Initial setup and maintenance demand time and resources

FUTURE IMPLEMENTATION

In the future, the Lost Item Logger System can be expanded with mobile app support for easier access, QR code tagging for quick identification of items, and AI-based image recognition to match uploaded photos with database records. Cloud integration may also be added to improve scalability and accessibility, while analytics could help identify common loss patterns and guide preventive measures.

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SYSTEM REQUIREMENTS

➢ SOFTWARE REQUIREMENTS:

The major software requirements of the project are as follows:
Language: Python.
Operating system: Windows 11.
Technical requirements: Monitor, Laptop, Keyboard, Mouse, etc...

➢ HARDWARE REQUIREMENTS:

The hardware requirements that map towards the software are as follows:

RAM: 8 GB.

Processor: i7.

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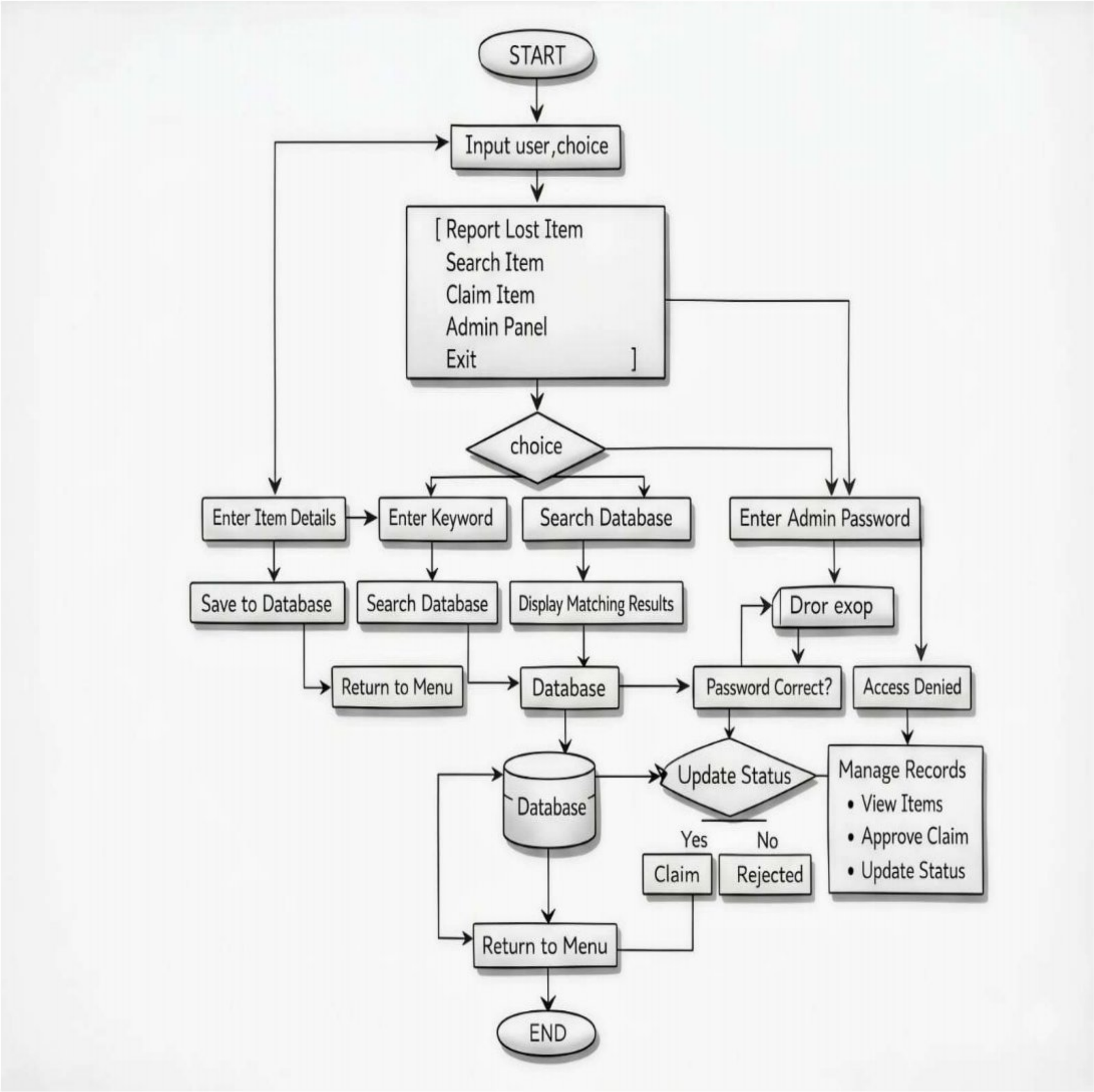
DATA FLOW DIAGRAM



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ALGORITHM

Step-1: Start.

Step-2: Read item name, category, location, date/time.

Step-3: Assign a unique log id.

Step-4: Display options.

1. Insert lost item .
2. Insert found item.
3. Search item.
4. View all records.

Step-5: Read user selection.

Step-6: If "Insert Lost item" →

1. Read item name.
2. Read category.
3. Read location.
4. Read date/time

Step-7: If "Insert Found item" →

1. Read item name.
2. Read category.
3. Read location
4. Read date/time

Step-8: If "Search item" →

1. Read search criteria
2. Check database for matches

Step-9: If match available → Allow user to claim item.

Step-10: Else → Print "Item not available."

Step 11: Admin verification process (approve/reject entries).

Step 12: Print item details (log ID, description, status).

Step 13: Stop

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IMPLEMENTATION

```
import csv
import os

LOST_FILE = "lost_items.csv"
FOUND_FILE = "found_items.csv"

# Initialize files if not exist
def initialize_files():
    if not os.path.exists(LOST_FILE):
        with open(LOST_FILE, 'w', newline='') as f:
            writer = csv.writer(f)
            writer.writerow(["ID", "Item Name", "Description", "Location", "Contact"])

    if not os.path.exists(FOUND_FILE):
        with open(FOUND_FILE, 'w', newline='') as f:
            writer = csv.writer(f)
            writer.writerow(["ID", "Item Name", "Description", "Location", "Contact"])

# Add lost item
def report_lost():
    item = input("Item Name: ")
    desc = input("Description: ")
    loc = input("Last Seen Location: ")
    contact = input("Contact Info: ")

    with open(LOST_FILE, 'a', newline='') as f:
        writer = csv.writer(f)
        writer.writerow([get_next_id(LOST_FILE), item, desc, loc, contact])

        print("Lost item reported successfully!")

# Add found item
def report_found():
    item = input("Item Name: ")
    desc = input("Description: ")
    loc = input("Found Location: ")
    contact = input("Contact Info: ")
```


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```
    with open(FOUND_FILE, 'a', newline='') as f:
        writer = csv.writer(f)
writer.writerow([get_next_id(FOUND_FILE), item, desc, loc, contact])

    print("Found item reported successfully!")

# View items
def view_items(file):
    with open(file, 'r') as f:
        reader = csv.reader(f)
        for row in reader:
            print(row)

# Get next ID
def get_next_id(file):
    try:
        with open(file, 'r') as f:
            return sum(1 for _ in f)
    except:
        return 1

# Match lost and found
def match_items():
    with open(LOST_FILE, 'r') as lost, open(FOUND_FILE, 'r') as found:
        lost_reader = list(csv.reader(lost))[1:]
        found_reader = list(csv.reader(found))[1:]

        print("\nPossible Matches:\n")
        for l in lost_reader:
            for f in found_reader:
                if l[1].lower() == f[1].lower():
                    print(f"Lost: {l}")
                    print(f"Found: {f}")
                    print("-" * 40)

# Menu
def menu():
    while True:
```


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```
    print("\n--- Lost Item Logger ---")
    print("1. Report Lost Item")
    print("2. Report Found Item")
    print("3. View Lost Items")
    print("4. View Found Items")
    print("5. Match Items")
    print("6. Exit")

    choice = input("Enter choice: ")
     if choice == '1':
        report_lost()
    elif choice == '2':
        report_found()
    elif choice == '3':
        view_items(LOST_FILE)
    elif choice == '4':
        view_items(FOUND_FILE)
    elif choice == '5':
        match_items()
    elif choice == '6':
        break
    else:
        print("Invalid choice!")

# Run program
initialize_files()
menu()
```


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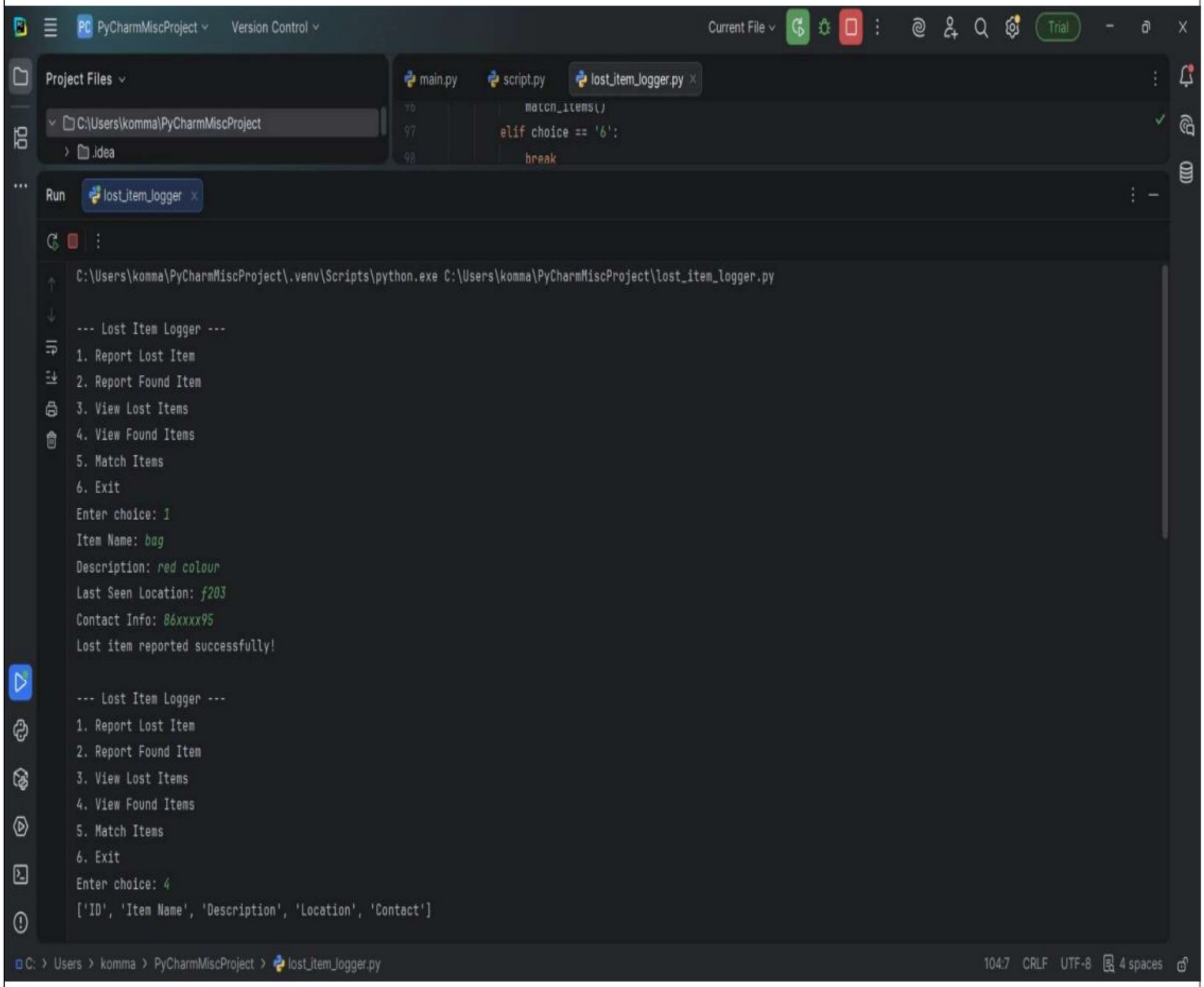
INTEGRATION AND SYSTEM TESTING OUTPUTS

Screen Shots:



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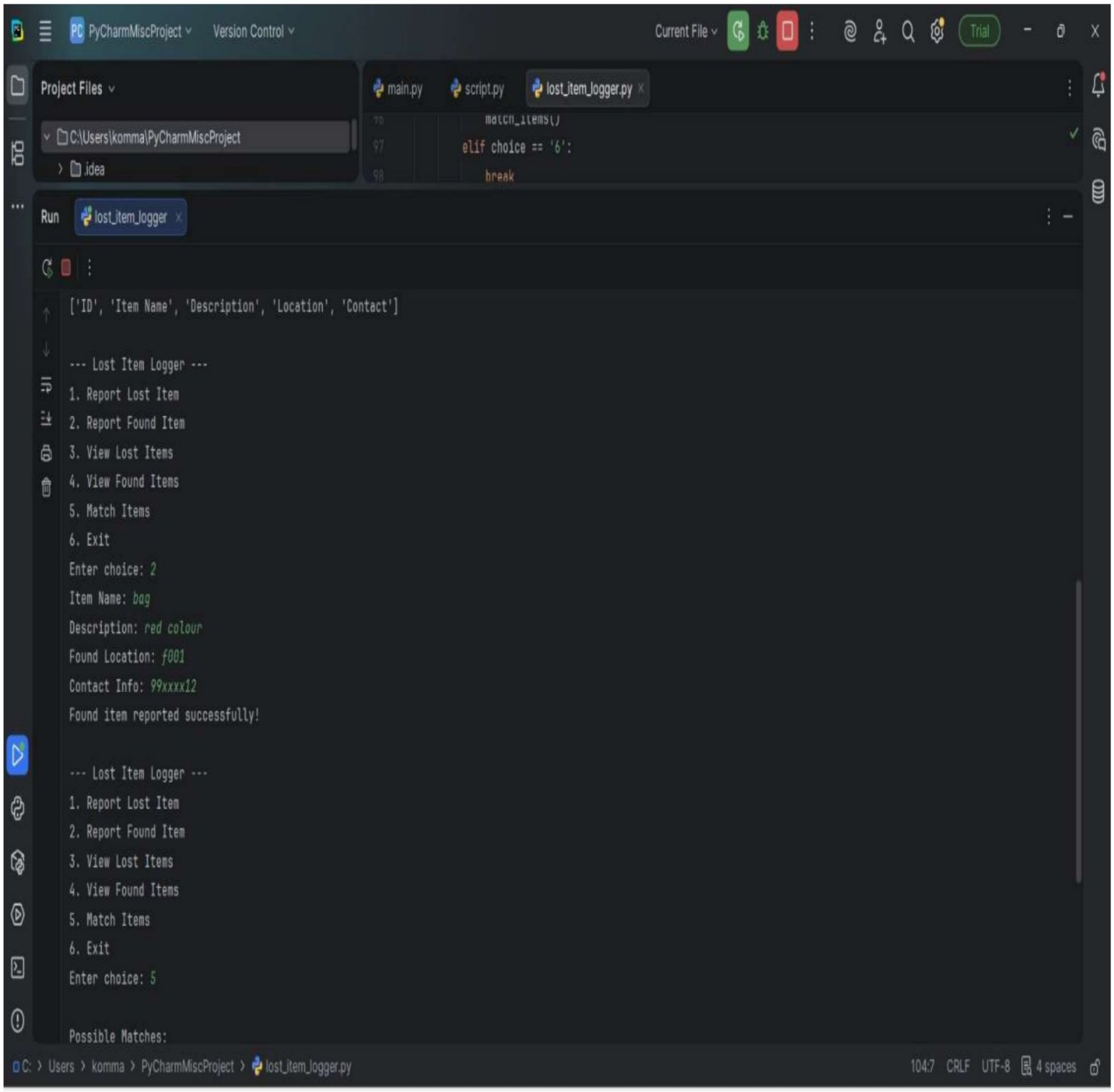

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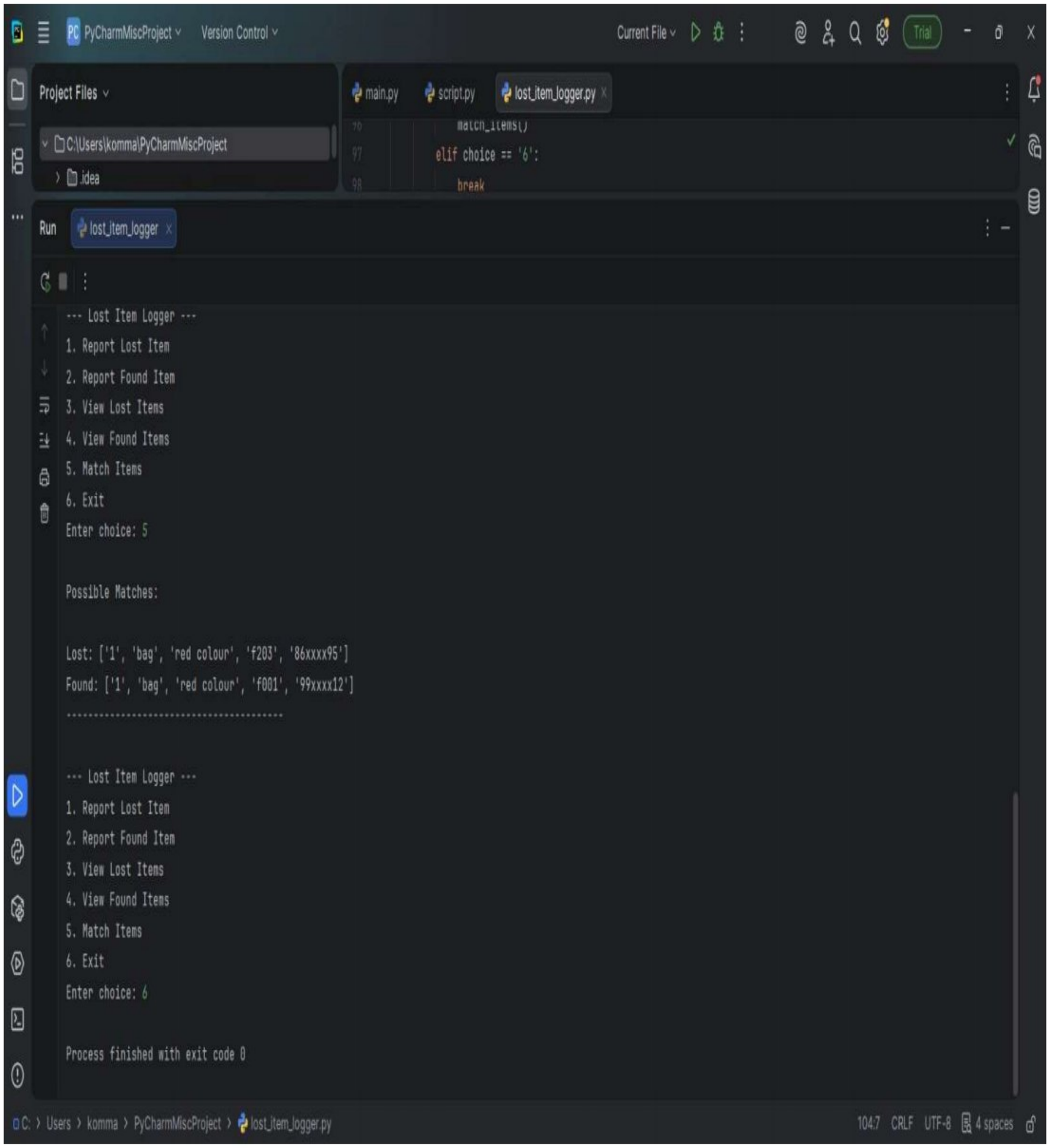

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CONCLUSION

The Lost Item Logger System offers a practical and organized way to handle lost-and-found situations. By allowing users to log items, track reports, and match them with rightful owners, it reduces confusion and saves valuable time. The system builds trust among users and provides a reliable framework that ensures belongings are not overlooked or forgotten.

Looking ahead, the system has the potential to grow with features like mobile alerts, image-based recognition, and cloud integration. These enhancements would make recovery faster and more accurate, while strengthening the sense of community and responsibility. In essence, the Lost Item Logger System is more than a tool—it is a step toward creating a supportive environment where lost items can be returned with ease and peace of mind.

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